

## Recurrent Neural Networks:

$$x = \{x_1, x_2, \dots, x_T\} \quad x_t \in \mathbb{R}^d$$

Batch size :  $b$

Sequence length :  $T$

Input feature dimension:  $d$

$h_{t-1} \in \mathbb{R}^h$  : hidden dimension.

$y \in \mathbb{R}^k$  : output dimension.

$$x \in \mathbb{R}^{T \times d}$$

Once the batch is considered

$$X \in \mathbb{R}^{b \times T \times d}.$$

RNNs:

$$h_t = \sigma(\omega_1 x_t + \omega_2 h_{t-1} + b_1) \in \mathbb{R}^h$$

$$\omega_1 \in \mathbb{R}^{h \times d}$$

$$\omega_2 \in \mathbb{R}^{h \times h}$$

$$b_1 \in \mathbb{R}^h$$

$$y_t = \omega_3 h_t + b_2$$

$$\omega_3 \in \mathbb{R}^{k \times h}$$

$$b_2 \in \mathbb{R}^k$$

## Gated Recurrent Unit: GRUs

$$h_t = \alpha_t \odot h_{t-1} + \beta_t \odot \tilde{h}_t$$

$$\alpha_t, \beta_t, h_{t-1}, \tilde{h}_t \in \mathbb{R}^h$$

$$\alpha_t = \sigma(w_\alpha [x_t; h_{t-1}] + b_\alpha)$$

$$[x_t; h_{t-1}] \in \mathbb{R}^{d+h}$$

$$w_\alpha \in \mathbb{R}^{h \times (d+h)}$$

$$b_\alpha \in \mathbb{R}^h$$

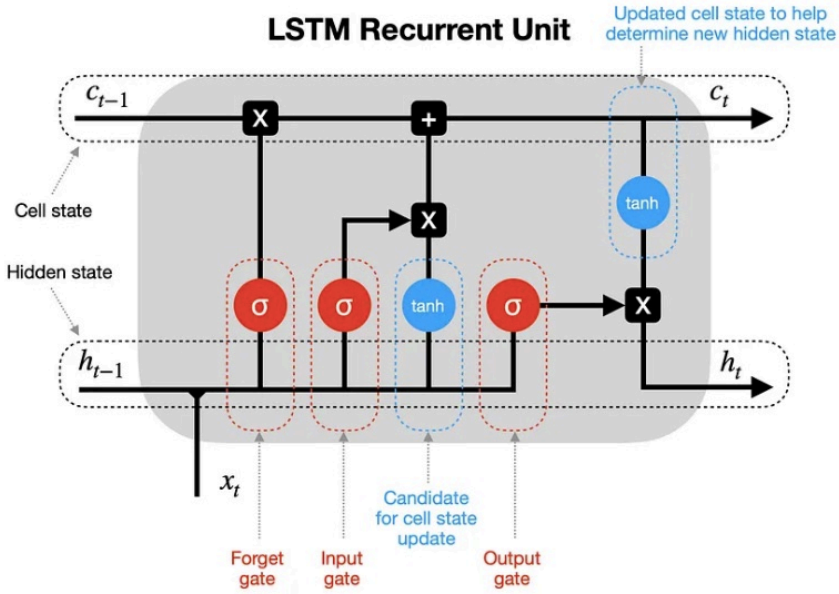
$$\gamma_t = \sigma(w_\gamma [x_t; h_{t-1}] + b_\gamma)$$

$$\tilde{h}_t = \sigma(w_h [x_t; \gamma_t \odot h_{t-1}] + b_h)$$

$$h_t = \alpha_t \odot h_{t-1} + (1 - \alpha_t) \odot \tilde{h}_t \in \mathbb{R}^h$$

$$y_t = w_3 h_t + b_3$$

# LSTMs: Long Short Term Memory



$$c_t, h_t \in \mathbb{R}^h$$

$$f_t = \sigma(w_f[x_t; h_{t-1}] + b_f)$$

$$i_t = \sigma(w_i[x_t; h_{t-1}] + b_i)$$

$$\tilde{c}_t = \tanh(w_c[x_t; h_{t-1}] + w_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$O_t = \sigma(w_o[x_t; h_{t-1}] + b_o)$$

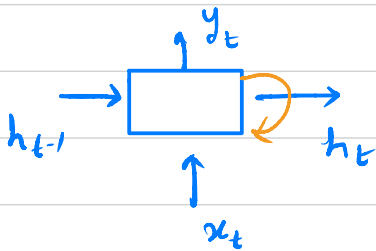
$$h_t = O_t \odot \tanh(c_t)$$

$$y_t = w_3 h_t + b_3$$

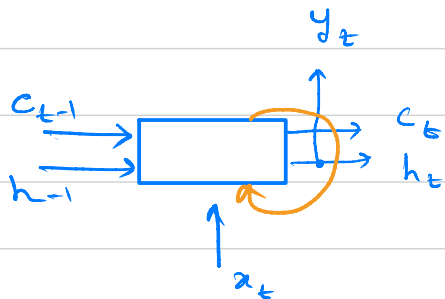
$$w_f, w_i, w_c, w_o \in \mathbb{R}^{h \times (d+h)}$$

$$b_f, b_i, b_c, b_o \in \mathbb{R}^h$$

$$w_3 \in \mathbb{R}^{k \times h}, b_3 \in \mathbb{R}^k$$



GRUs / RNNs.



LSTMs.

